

Assignment No. 2

Git and Git Commands

Aim :

To study and perform basic Git commands for managing a project and uploading it to GitHub.

Introduction

Git is a version control system used to track changes in project files. GitHub is an online platform used to store and manage Git repositories. Git commands help us initialize a repository, track files, create commits, and upload the project to GitHub.

1. git init :

Theory:

Initializes a Git repository in the current project folder. It allows Git to track changes in the project.

Command:

git init

```
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Dell\OneDrive\Desktop\vsec_git_commands>git init
Initialized empty Git repository in C:/Users/Dell/OneDrive/Desktop/vsec_git_commands/.git/
```

2. git status :

Theory:

Displays the current status of the Git repository and shows files that are new, modified, or ready to be committed.

Command:

git status

```
C:\Users\Dell\OneDrive\Desktop\vsec_git_commands>git status
On branch main

No commits yet

nothing to commit (create/copy files and use "git add" to track)
```

3. git add . :**Theory:**

Adds all new and modified files to the staging area before committing.

Command:

git add .

```
C:\Users\Dell\OneDrive\Desktop\vsec_git_commands>git add .

C:\Users\Dell\OneDrive\Desktop\vsec_git_commands>git commit -m "First commit"
[main (root-commit) e2eb1cb] First commit
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 assign_1/index.html

C:\Users\Dell\OneDrive\Desktop\vsec_git_commands>
```

4. git config --list :

Theory:

Displays the configuration settings of Git, such as username and email.

Command:

git config --list

```
C:\Users\Dell\OneDrive\Desktop\vsec_git_commands>git config --list
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=openssl
http.sslcainfo=C:/Program Files/Git/mingw64/etc/ssl/certs/ca-bundle.crt
core.autocrlf=true
core.fscache=true
core.symlinks=false
pull.rebase=false
credential.helper=manager
credential.https://dev.azure.com.usehttppath=true
init.defaultbranch=main
user.name=Sanskruti Jadhav
user.email=sanskrutijadhav186@gmail.com
credential.helper=
credential.helper=C:/Program Files/Git/mingw64/bin/git-credential-manager.exe
core.repositoryformatversion=0
core.filemode=false
core.bare=false
core.logallrefupdates=true
core.symlinks=false
core.ignorecase=true
```

5. git config user.email :**Theory:**

Sets the email address used to identify the user in Git commits.

Command:

git config user.email "your_email"

```
C:\Users\Dell\OneDrive\Desktop\vsec_git_commands>git config user.email "sanskrutijadhav186@gmail.com"

C:\Users\Dell\OneDrive\Desktop\vsec_git_commands>git add .

C:\Users\Dell\OneDrive\Desktop\vsec_git_commands>git status
On branch main
nothing to commit, working tree clean

C:\Users\Dell\OneDrive\Desktop\vsec_git_commands>
```

6. git remote add origin :**Theory:**

Connects the local Git repository to a remote GitHub repository.

Command:

git remote add origin <repository-url>

```
C:\Users\Dell\OneDrive\Desktop\vsec_git_commands>git remote add origin https://github.com/sanskrutijadhav186-cmd/VSEC-ASSIGNMENTS.git
error: remote origin already exists.

C:\Users\Dell\OneDrive\Desktop\vsec_git_commands>git remote -v
origin https://github.com/sanskrutijadhav186-cmd/VSEC-ASSIGNMENTS.git (fetch)
origin https://github.com/sanskrutijadhav186-cmd/VSEC-ASSIGNMENTS.git (push)

C:\Users\Dell\OneDrive\Desktop\vsec_git_commands>
```

7. git remote -v :

Theory:

Displays the remote repository URL connected to the local Git repository.

Command:

git remote -v

```
C:\Users\Dell\OneDrive\Desktop\vsec_git_commands>git remote add origin https://github.com/sanskrutijadhav186-cmd/VSEC-ASSIGNMENTS.git
error: remote origin already exists.

C:\Users\Dell\OneDrive\Desktop\vsec_git_commands>git remote -v
origin https://github.com/sanskrutijadhav186-cmd/VSEC-ASSIGNMENTS.git (fetch)
origin https://github.com/sanskrutijadhav186-cmd/VSEC-ASSIGNMENTS.git (push)

C:\Users\Dell\OneDrive\Desktop\vsec_git_commands>
```

8. git commit -m :**Theory:**

Saves the staged changes in the local repository with a meaningful message.

Command:

git commit -m "uploaded"

```
C:\Users\Dell\OneDrive\Desktop\vsec_git_commands>git commit -m "uploaded"
On branch main
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   assign_1/index.html

no changes added to commit (use "git add" and/or "git commit -a")

C:\Users\Dell\OneDrive\Desktop\vsec_git_commands>
```


9. git branch -M main :

Theory:

Renames the current branch to main.

Command:

git branch -M main

```
C:\Users\Dell\OneDrive\Desktop\vsec_git_commands>git branch
* main
```

10. git push -u origin main :

Theory:

Uploads the committed project from the local repository to the main branch on GitHub.

Command:

git push -u origin main

```
C:\Users\Dell\OneDrive\Desktop\vsec_git_commands>git push -u origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Writing objects: 100% (4/4), 265 bytes | 44.00 KiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/sanskrutijadhav186-cmd/VSEC-ASSIGNMENTS.git
* [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```

11. git push :

Theory:

Uploads the latest committed changes to the connected GitHub repository.

Result :

The basic Git commands were successfully performed and the project was connected and uploaded to GitHub.

Conclusion :

Git provides an easy way to track project changes and maintain versions, while GitHub allows the project to be stored and managed online.